



MLOps & Model Deployment — What to Expect

A hands-on, 3-day workshop for UG students (CSE · AI & DS · IT)

Over three days you'll take **one machine-learning model all the way to a live, monitored deployment** — train it, track it, wrap it in an API, containerize it, automate it, push it to a public URL, and watch it for drift. This is a build-along workshop. We run everything in **GitHub Codespaces** — a free, ready-made coding environment in your browser — so there is **nothing to install**. By day 3 you'll have a working model live on the internet you can put on your CV.

DURATION 3 continuous days · 5 hrs/day	DATES 6 – 8 August 2026 (Thu–Sat)	VENUE University campus — A/C classroom + lab
AUDIENCE UG · CSE / AI & DS / IT (up to 30)	PREREQUISITES Basic Python; intro ML	COST TO YOU Free tools only

What you'll learn

Understand the ML lifecycle and why models fail after training; track experiments and version models (MLflow); package and serve a model as a REST API (FastAPI); containerize and automate it (Docker, GitHub Actions); deploy it to a free public host; and monitor it for drift with a plan to retrain.

The pipeline you'll build (one project, all 3 days)

1	Train & track	Train a model and log every run to MLflow; register & version the best one.
2	Serialize & serve	Freeze the model and expose it via a FastAPI REST API with input validation.
3	Containerize	Package the API in a Docker image that runs anywhere.
4	Automate (CI/CD)	Lint, test and build on every push with GitHub Actions — plus a model accuracy gate.
5	Deploy	Ship the container to a free host (Hugging Face Spaces / Render) — a public URL.
6	Monitor	Detect data drift, alert, and define a retraining trigger.

How it runs & how you're graded

Hands-on & interactive. ~70% of class time is live-coding and labs. You'll pair up, take on "break it" challenges, and finish on a demo day.

ASSESSMENT A short **quiz**, a **live demo** of your deployed model, and a **mock technical interview** with the industry expert.

How to prepare — before Day 1

DO THIS FIRST — EVERYONE, NO EXCEPTIONS

- **Create two free accounts and verify the confirmation email for each:** **GitHub** (github.com) and **Hugging Face** (huggingface.co). No card needed. **Unverified accounts cannot be used in the labs** — and signing up in class wastes everyone's time.
- **Bring your laptop and charger.** Basic Python & intro ML is enough — you get starter code and wire the pieces together.

Then choose how you'll run the labs. Both are fully supported and every worksheet gives commands for each.

OPTION B · CODESPACES

RECOMMENDED

Nothing to install — ~3 minutes on the day. A full Linux machine in your browser. Just make sure your GitHub account is verified.

1. Open `github.com/anandvenkat4/mlops-sample-project`
2. Code ▾ → **Codespaces** → **Create codespace on main**
3. Wait ~3 min — it installs everything and trains a model for you
4. Verify: `bash .devcontainer/check.sh`

OPTION A · YOUR OWN LAPTOP

Allow 45 minutes, and finish before Day 1. Needs 8 GB RAM and ~10 GB free disk.

1. Install **Python 3.11** from python.org. **Windows:** tick **"Add python.exe to PATH"**.
2. Install **Git** (git-scm.com) and **Docker Desktop** (docker.com) — Docker is needed from Day 2
3. `git clone https://github.com/anandvenkat4/mlops-sample-project` then `cd mlops-sample-project`
4. `python -m venv .venv` then activate it (`source .venv/bin/activate`, Windows `.venv\Scripts\activate`)
5. `pip install -r requirements.txt`
6. Verify: `python make_data.py`, `python train.py`, `pytest -q`, `docker run hello-world`

⚠ **Laptop users** — the one detail that breaks most setups: your Python must be **3.10, 3.11 or 3.12**. **Python 3.13 or newer will not work** (our pinned `numpy==1.26.4` has no build for it, and `pip install` fails outright). Run `python --version` before installing anything else. Stuck? Use Codespaces — you can switch at any time, even mid-workshop, and lose nothing.